

Deliverable 2

Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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This deliverable reports how the plan set out in Deliverable 1 was carried out: how the evidence on structural aliasing was collected from the retrained Chronos-Bolt sweep, how it was turned into a Bayesian analysis, and what the analysis found.

Data Collection and Probing

Models and signals

We analyse the five geometries of Deliverable 1: $P=S=16$, the shorter strides $S=12$ and $S=8$ at $P=16$, and $P=S=8$ and $P=S=24$. Four of them were trained with the same 100k-step budget, so they can be compared directly. The $P=S=16$ run did not complete correctly and is being repeated. We show it for continuity but leave it out of every comparison between geometries.

Each context is 480 samples at $f_s=512$ Hz, with a 64-step horizon, over $[2, 250]$ Hz. We use 480 because it is divisible by every stride in the sweep, so the patch grid closes exactly. The 512 of the probing convention would leave $S=12$ and $S=24$ with a padded last patch, whose artefact cannot be told apart from the effect we are testing.

As described in Deliverable 1, we use 100 TSMixup and 100 KernelSynth signals. Each is normalised to unit variance and carries an injected tone with a known frequency and phase: the corpus gives realism, the tone gives the known quantity. The sine set is kept as the pure-sinusoid mode of the collapse sweep.

What enters the inference

Two filters sit between the collected measurements and the models, and both cost data. The first keeps a lock only when *both* of its controls $f_k \pm 0.25f_s/S$ fall off $\mathcal{F}_{lock}$; a lock whose control would land on another member is dropped whole, since that would compare a lock against a lock rather than against a clean neighbour. The paired contrast of Equation 1 is then applied unchanged to the survivors, so the estimator is exactly the one preregistered in Deliverable 1 — only the set of admissible sites is narrowed. This bites only at $P=16, S=12$, where the patch grid (32 Hz spacing) and the stride grid (42.6 Hz spacing) interleave so that an offset of $0.25f_s/S$ reaches the neighbouring lock exactly; it leaves the 3 usable sites $\{64, 128, 192\}$ Hz out of 11. These survivors are patch-grid or coincident sites: at this offset the geometry’s own stride locks (42.6, 85.3, ... Hz) are not $H1$ -testable, and recovering them would require a one-sided control. The second

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

drops a contrast whose recovery is near zero at the lock *and* at both controls, because that is a ratio of two noise floors and says nothing about $H1$. [TO FILL, fraction of collected contrasts removed by this live filter, overall and per geometry] Both filters are limits of the present design rather than choices, and they are why some geometries weigh less than others in $\bar{\beta}$. As a check on the design itself, the collection stops if either level of the lock indicator is empty, because the effect would then not be identified and the posterior would only repeat the prior.

The sine sweep

Figure 1 and Figure 2 show the sine sweep of Deliverable 1, grouped by patch length. In each row, the left panel gives the frequency the model generates against the frequency it was given, and the right panel gives whether that frequency can still be read from the representation. Both views get worse towards the top of the band, and they do not agree with each other: a tone can still be readable and yet be rebuilt as the wrong frequency.

What the measurements show

Three tables summarise the empirical core, each computed directly from the models by `make_tables.py`. Table 1 gives the phase-lock frequency families and the stride mitigation: the full token collapse sits on the stride grid $c f_s/S$, and as S shrinks its lowest member f_s/S rises ($32 \rightarrow 42.6 \rightarrow 64$ Hz) with fewer members in band. Table 2 reads the forecast of the official model tone by tone: the octave-aligned c_{pp} 1, 2, 4 are rebuilt while $c_{pp}=3$ (96 Hz) is aliased onto 128 Hz — the one genuine blind spot. Table 3 collects the hypothesis verdicts across geometries: $H1$ is refuted in its strong form (the lock is rebuilt, mean contrast $d > 0$), $H2$ holds (small phase CV), and $H3$ holds (the across-patch token dispersion vanishes at the predicted locks).

Model	f_s/S (Hz)	#locks	stride locks $c f_s/S$ (Hz)
p16-s16	32	7	32 64 96 128 160 192 224
p16-s12	42.7	5	42.67 85.33 128 170.67 213.33
p16-s8	64	3	64 128 192
p8-s8	64	3	64 128 192
p24-s24	21.3	11	21.33 42.67 64 85.33 106.67 128 149.33 170.67 192 213.33 234.67

Table 1: Phase-lock frequency families per geometry ($f_s = 512$ Hz, band $[2, 250]$ Hz). The token collapse sits on the stride grid $c f_s/S$, whose lowest member is f_s/S . As the stride S shrinks, f_s/S rises and fewer members fall in band — the stride mitigation. (A weaker anti-periodic dip also sits at the half-stride points $(c + \frac{1}{2})f_s/S$.)

c _{pp}	f (Hz)	recovery R	out dom (Hz)	verdict
0.5	16	0.915	16	faithful
1	32	0.923	32	faithful
1.5	48	0.087	40	decays
2	64	0.821	64	faithful
2.5	80	0.104	84	decays
3	96	0.017	128	ALIASED
3.5	112	0.02	116	decays
4	128	0.974	128	faithful

Table 2: Reconstruction on the official *chronos-bolt-tiny* ($P=S=16$): a pure tone is swept and its forecast measured. Octave-aligned c_{pp} (1,2,4; period divides P) are rebuilt; $c_{pp}=3$ (**96 Hz**) is aliased onto **128 Hz**; half-integer c_{pp} decay. ‘out dom’ is the dominant output frequency.

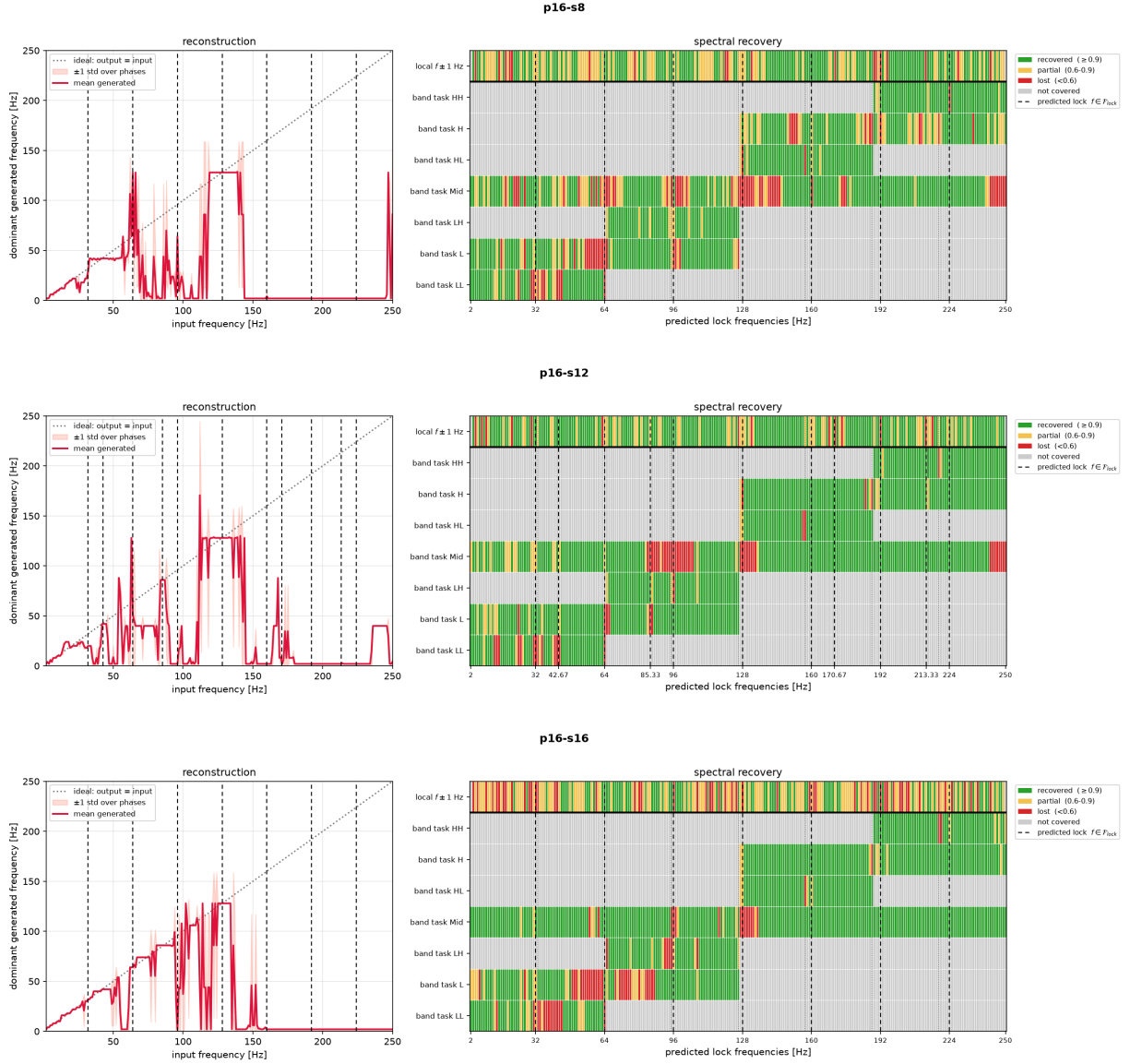


Figure 1: The three $P=16$ geometries under a pure-sinusoid sweep at 1 Hz over $[2, 250]$ Hz. The stride gets shorter from top to bottom, while the patch grid stays the same. Each row shows the same sweep in two ways. The left panel plots the frequency the model generates against the frequency it was given: a point on the diagonal is a tone rebuilt correctly, and a flat stretch means that several inputs give the same output. The right panel asks instead whether the input frequency can still be read back from the model's representation, with one row per probing task. Green means the frequency is recovered, red means it is lost, and grey means that the task does not cover it. The dashed lines mark the geometry's own phase-lock set F_{lock} . Both panels get worse towards the top of the band, but they do not fail at the same frequencies.

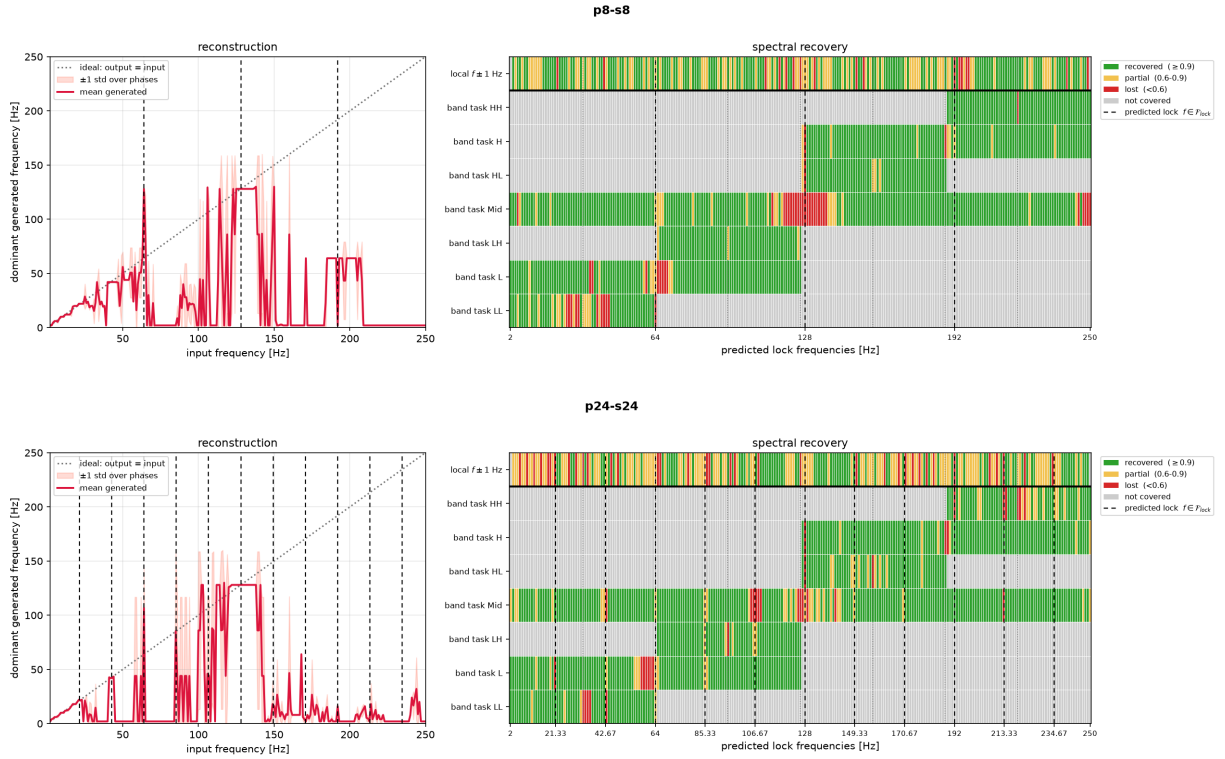


Figure 2: The two geometries where the patch and the stride are equal, $P=S=8$ and $P=S=24$, drawn and read as in Figure 1. When $P=S$, the patch grid kf_s/P and the stride grid cf_s/S fall on the same frequencies, so one set of dashed lines covers both. These two geometries therefore cannot tell which of the two grids a dip belongs to. Only the $S < P$ rows of Figure 1 can do that, and this is why the location model of H3 depends on them.

Model	H1	mean d	H2	CV	H3	std@lock
p16-s16	REFUTED	1.04	SUPPORTED	0.08	PASS	0
p16-s12	REFUTED	2.55	SUPPORTED	0.14	PASS	0
p16-s8	REFUTED	2.06	SUPPORTED	0.21	PASS	0
p8-s8	REFUTED	3.59	SUPPORTED	0.12	PASS	0
p24-s24	REFUTED	2.4	SUPPORTED	0.09	PASS	0

Table 3: Hypothesis verdicts across geometries (pure sinusoid). **H1 refuted** (mean log-contrast $d > 0$: the lock is rebuilt at least as well as its neighbours); **H2 holds** (small phase CV); **H3 holds** (the across-patch token std at the predicted locks is ≈ 0). Verdicts are unchanged under a TSMixup background (see the testing README).

Model	Hypothesis	Estimand	Input
A	H1 behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	contrasts
B	H1 representational	$e^{\theta_{lock}}$: codelength at a lock / elsewhere	mdl_cells
C	H2	σ_ϕ : spread of the deficit across phase	contrasts
D1	H3 location	which grid the sites lie on (LOO)	collapse
D2	H3 movement	κ_S : slope of measured on predicted spacing	sites

Table 4: The five regressions that make up the analysis. For each one the table gives the hypothesis it addresses, the quantity whose posterior decides that hypothesis, and the table from the probing stage it is fitted to. They are not five views of the same measurement: H1 is tested twice, once on what the model forecasts (A) and once on what its representation encodes (B), while H3 is split into a question of location, decided by model comparison rather than by a single coefficient (D1), and a question of movement (D2). Models A and C do share an input, but not a response: A uses the phase-averaged contrast d of Deliverable 1, and C uses the same measurement with the phase index kept.

Bayesian Analysis Design

Deliverable 1 applied a two-part framework to H1 only. Here each hypothesis gets its own model, and the five models are listed in Table 4. All five are ordinary regressions. What changes between them is the quantity on the left and the question the coefficient answers. The priors in Table 5 are unchanged from Deliverable 1, and every effect prior is centred on *no effect*, so a finding cannot come from the prior.

Reading the models

Every model below has the same two parts. A *linear predictor* μ_i says what value we expect the measurement to take, and it is built by adding one offset for each group the observation belongs to. A *likelihood* says how far a single observation is allowed to fall from that expected value. The quantity that decides a hypothesis is always one coefficient of the predictor. The other terms are there so that this coefficient is not confounded by the design.

H1: how large is the effect

The measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$. Both amplitudes are fitted by least squares at the known frequency, over the same 64-step horizon, from the median forecast. We collect it in matched triplets: each lock f_k with controls at $f_k \pm 0.25f_s/S$, all three sharing the same background and phase. This is what makes the contrast paired. The contrast is unchanged from Deliverable 1:

$$d = \log(R_k + 0.01) - \frac{1}{2} [\log(R_- + 0.01) + \log(R_+ + 0.01)], \quad (1)$$

one number per measured triplet, negative when the lock recovers *less* than its neighbours. It can be formed only because the injected tone has a known frequency: Equation 1 cannot be built from a generated signal on its own. Model A fits it as

Model	Parameter	Prior	Role
A, C	$\bar{\beta}$	Student- $t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student- $t_4(0, 0.5)$	slope of the effect in patch overlap $O = (P - S)/P$
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student- $t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half- $\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar{y}, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	k	Gamma(2, 0.1)	Gamma shape (dispersion)
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry level and the two grid labels; $H3$ predicts $\theta_S < 0$
D2	κ_0	$\mathcal{N}(0, 10)$ Hz	offset of the spacing regression; $H3$ predicts 0
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	slopes on f_s/S and f_s/P ; $H3$ predicts 1 and 0

Table 5: The prior on every parameter of the five models, unchanged from Deliverable 1, together with the role each parameter plays in its regression. Every effect prior is centred on no effect: zero for the contrast and codelength coefficients, no dependence on phase for σ_ϕ , and no relationship between measured and predicted spacing for κ_S . Each hypothesis therefore has to come from the data rather than from the prior, and a posterior that stays where its prior was is itself a report of an absent effect. Scale parameters are given half- t priors on the same units as the response they disperse. The two $H3$ rows also give the values the hypothesis predicts, which is what the posteriors will be compared against.

$$d_i \sim \text{Student-}t_4(\mu_i, \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}(\bar{\beta} + \delta_O \tilde{O}_c, \tau). \quad (2)$$

- β_c : the typical contrast for configuration c , one value per geometry.
- u_k : one offset per lock harmonic, so that an unusual frequency does not distort the geometry it belongs to.
- u_b : one offset per background draw, one level for each of the 200 TSMixup and KernelSynth signals. If a result holds for only one of the two corpora, σ_b comes out large.
- $\bar{\beta}$: the average contrast across configurations. **This is the estimand for $H1$** , and $e^{\bar{\beta}}$ reads as how much recovery changes at a lock compared with its controls.
- δ_O : the slope of β_c in the patch overlap $\tilde{O}_c$, with $O_c = (P_c - S_c)/P_c$. It tests the mitigation proposed in Deliverable 1: if overlapping the patches reduces the effect, it shows up here.
- τ : how much the configurations scatter around $\bar{\beta}$.
- σ and the Student- t_4 : the residual scale and the shape of the noise. The heavy tails matter because recovery falls to near zero where the model rebuilds nothing, and the logarithm then gives very large contrasts. Under a Gaussian, a few of those would set $\bar{\beta}$; under t_4 they are tolerated.

Writing β_c as a draw around $\bar{\beta}$ makes each geometry count as one unit. The number of observations per geometry varies several times over with how many lock sites fall inside the band, so a plain average over all rows would measure the design instead of the effect.

Model B asks the same question of the representation rather than of the forecast. Its measurement is a *frequency-local* codelength. At each candidate frequency f_c , a probe reads the 256-d [REG] vector at ten points, one per encoder and decoder block plus the pre-projection vector and the quantile head, and must separate tones at $f_c \pm 1$ Hz. The estimator is prequential: each chunk of labels is encoded by a probe trained

only on the chunks before it, so a probe that memorises noise pays for itself and gains nothing. The seven band tasks of Deliverable 1 span the whole band, so a single codelength would summarise hundreds of frequencies at once, leaving the lock indicator nothing to attach to. The regression is Gamma with a log link on the codelength y_i :

- α_0 : the baseline log codelength away from a lock.
- θ_{lock} : the log expansion at a locked frequency, so that $e^{\theta_{lock}}$ reads as how many times more bits a locked frequency costs. **This is the estimand.**
- k : the Gamma shape, which absorbs the dispersion of a quantity that is positive and right-skewed.

H2: does the phase matter

This is the first of the two extensions, on the same triplets as H1. Equation 1 has already averaged over phase, so it cannot answer a question about phase. Here we keep the phase of each measurement, cut the phase circle into eight slices, and give each slice its own offset:

$$\mathbb{E}[y_i] = \beta_{c[i]} + u_{k[i]} + u_{p[i]}, \quad u_p \sim \mathcal{N}(0, \sigma_\phi). \quad (3)$$

- y_i : the per-phase deficit. It is the same measurement as d , but with the phase index kept instead of averaged away.
- u_p : one offset per phase slice, eight in total.
- σ_ϕ : the spread of those eight offsets. **This is the estimand for H2**, and it answers the question with a single number. If the deficit depends on where in its cycle the signal starts, the offsets differ and σ_ϕ is large. If the geometry alone decides it, the offsets are about equal and σ_ϕ is close to zero.
- β_c, u_k : as in Model A, so that the phase term does not absorb differences between geometries or harmonics.

H2 predicts $\sigma_\phi \approx 0$. We use slices rather than a fitted curve because slices assume nothing about the *shape* of a possible dependence: any systematic variation with phase makes the offsets differ.

H3: where the dips are, and whether they move

This is the second extension. It needs two models because H3 makes two claims. Both are fitted to the token collapse $z_g(f)$, the dispersion of the input-patch embeddings across patches, which is zero exactly when consecutive patches are identical. We sweep it on one shared grid: a uniform 1 Hz sweep joined with the predicted sites of *every* configuration. In this way no geometry is scored only where its own theory predicts a dip, and non-integer sites such as 42.6 Hz for $S=12$ are covered.

Which grid are the dips on? Every swept frequency carries two yes/no labels: whether it lies on the stride grid $\{cf_s/S\}$, and whether it lies on the patch grid $\{kf_s/P\}$:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid}, \sigma). \quad (4)$$

- $z_g(f)$: the dispersion of the patch embeddings across patches, at frequency f for geometry g . We take the logarithm so that the two labels act multiplicatively.
- α_g : the off-grid level for that geometry, which differs between models and is not of interest here.
- θ_S : how much lower the dispersion is on the stride grid than off it. **This is the estimand**, and H3 predicts $\theta_S < 0$.
- θ_P : the same for the patch grid, so that the two families compete instead of one being assumed.

We also fit each label on its own, and once with neither, which shows which grid explains the dips. Only the $S < P$ geometries can settle this, because at $P=S$ the two grids fall on the same frequencies.

Do the dips move with the geometry? A good fit above is still compatible with dips that only happen to coincide with a fixed grid. We therefore regress the measured spacing on the two predicted spacings:

$$\hat{\Delta}_g \sim \mathcal{N}\left(\kappa_0 + \kappa_S \frac{f_s}{S_g} + \kappa_P \frac{f_s}{P_g}, \sigma\right). \quad (5)$$

- $\hat{\Delta}_g$: the spacing read off the detected dips, without assuming what produced them.
- κ_S : the slope on the spacing predicted by the stride. **This is the estimand**, and *H3* predicts exactly 1.
- κ_P : the slope on the spacing predicted by the patch, which *H3* predicts to be 0.
- κ_0 : an offset, predicted to be 0.

There are two caveats. First, P varies only along $P=S$, so κ_P is confounded and the series at fixed $P=16$ carries the answer. Second, we measure the spacing in two ways, as the lowest dip and as the median gap, and fit both, because one spurious detection halves the median while the lowest dip is unaffected.

Decisions and validation

The thresholds are fixed before we see the posterior: the probability of at least 20% attenuation, $\Pr(\bar{\beta} < \log 0.8 \mid D)$, and the equivalence regions $|\bar{\beta}| < \log 1.1$, $\sigma_\phi < \log 1.1$ and $|\kappa_S - 1| < 0.1$. A hypothesis is supported at probability ≥ 0.95 , refuted at ≤ 0.05 , and *inconclusive* in between. A pass/fail test cannot give that third outcome, which is often the honest answer here. Competing formulations are compared by leave-one-out cross-validation.

We validate before we interpret. Prior predictive checks come before collection. A recovery check then simulates data with a known effect and confirms that the model finds it, which separates “no effect” from “a design that cannot see one”. Sampling uses NUTS, and we report a fit only when $\hat{R} < 1.01$, the effective sample sizes are above 1000, and there are no divergences.

Results

[TO FILL, Written once the notebook completes. Introductory summary, approximately 500 words, one short paragraph per marker.]

Collected evidence

[TO FILL, Size and balance of the collected design: contrast observations per geometry and how many are live, MDL cells split by lock indicator, swept frequencies and replicates. State that the prior probability of each decision threshold is reported next to the posterior, since a posterior probability that has not moved from its prior is not evidence, and that the parameter-recovery check was passed, which is what separates “no effect” from “a design that cannot see one”.]

H1

[TO FILL, Model A: $e^{\bar{\beta}}$ with its interval and $\Pr(\bar{\beta} < \log 0.8 \mid D)$, the sign of the effect, and whether the overlap slope δ_O supports the mitigation. Model B: $e^{\theta_{lock}}$ and whether the representational reading agrees with the behavioural one. One sentence on the comparison with the earlier frequentist result, which refuted *H1* in its strong form.]

H2

[TO FILL, Model C: σ_ϕ with its interval, $\Pr(\sigma_\phi < \log 1.1 \mid D)$, and the leave-one-out comparison against the phase-free model. If the magnitude and the evidential readings disagree, report both rather than collapsing them into one verdict.]

H3

[TO FILL, Model D1: which grid wins by leave-one-out, in each signal mode, and the size of θ_S . Model D2: the slope κ_S against the equivalence region around 1, with the fixed- $P=16$ series as the identifying fit and κ_P reported as confounded.]

Verdicts and limitations

[TO FILL, One table: hypothesis, estimand, posterior summary, decision rule, probability, verdict. Then the limitations in two or three sentences: convergence against the preregistered thresholds, one training seed per geometry, the $P=S=16$ training still to be repeated, and the reduced budget behind the frequencies where nothing is recovered.]